

## Is Cybersecurity Safe From AI?

The already-in-it guide — the full scores by track, the six factors behind them, and what to do from where you already are.

5.0

AI EXPOSURE · INCIDENT RESPONSE / THREAT HUNTING

where 10 is most at risk

For anyone already in it — in the degree, or job-hunting.



## Before you read this

The choice is already made — you're in a cybersecurity degree or training, or out of one and into the job hunt. So this guide skips the "should you study cybersecurity" question and goes straight to the one that's actually live: given where you already are, what do you do about it?

It's written to you directly — whether you are the one in it, or a parent reading on their behalf. Where it says "you," a parent can simply read "them"; the moves are the same either way.

**What this is.** A facts document, not a pep talk. It sets out what's known about where cybersecurity sits in an AI economy, where the evidence is contested, and where we might be wrong — and then what to actually do from here.

**The good news, up front.** Already in it is not too late — not close. Cybersecurity isn't one destination, it's a wide spread of them, from 5.0 to 7.4, and almost none of your exposure is fixed yet. Where you land inside that range is decided mostly by moves still in front of you — chiefly, whether you steer toward the roles that face the adversary directly. The second half of this guide is those moves.

**And one thing specific to cybersecurity.** This is the field with the largest gap between how it is marketed and how it actually hires. You've almost certainly met the figure of 4.8 million unfilled cybersecurity jobs — in prospectuses, bootcamp advertising and certification marketing everywhere. The number is real. What it means for someone at the start of a career is not what those sources imply, and the section on that figure explains why. If you picked cybersecurity because it was sold as the safe tech career, that gap is the most important thing in this document to understand now.

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## The short answer

**Better than most technical paths, and not as protected as the marketing suggests.**

Incident response and threat hunting scores 5.0 out of 10. Tier-one SOC analyst work scores 7.4.

Cybersecurity is the best-scoring pure-technology track we measure — but "safer than software" and "safe" are different claims. Its floor is 5.0, against 4.0 for licensed engineering and 2.5 for a service electrician.

And the entry route everyone is pointed toward is the specific part being automated — which is why, from where you already are, the work now is climbing toward the protected end, not just landing anywhere.

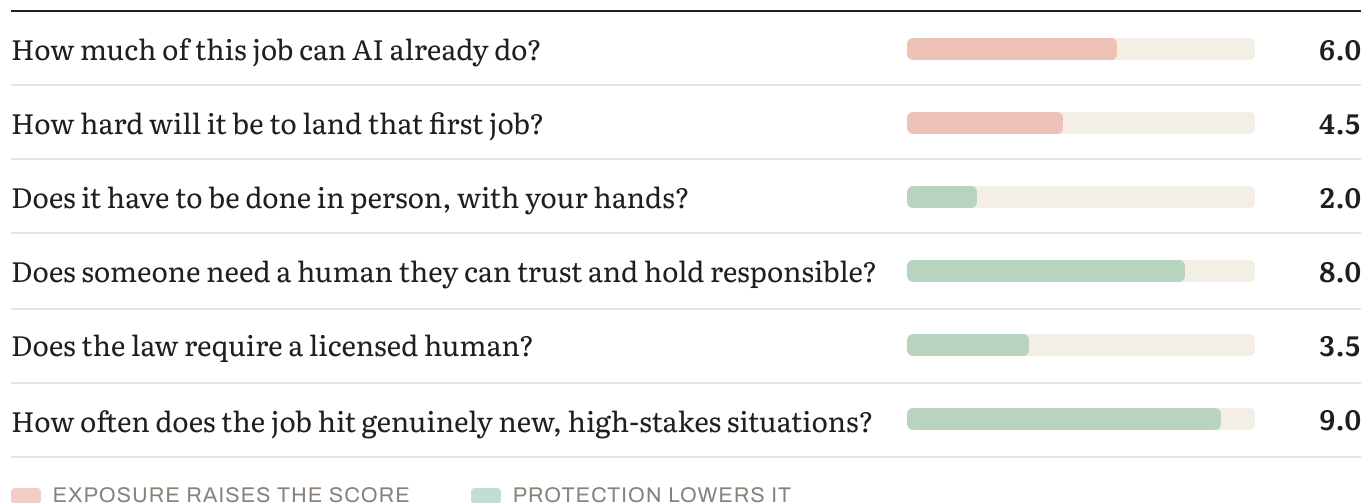
## The scores

| ROLE                               | 2023 | 2025 | FALL 2026 | 3-YR MOVEMENT | BAND          |
|------------------------------------|------|------|-----------|---------------|---------------|
| Incident response / threat hunting | 4.2  | 4.6  | 5.0       | +0.8          | Moderate      |
| Security architecture              | 4.7  | 5.1  | 5.5       | +0.8          | Moderate      |
| Penetration testing / offensive    | 4.9  | 5.3  | 5.7       | +0.8          | Moderate      |
| GRC / compliance                   | 5.2  | 5.7  | 6.1       | +0.9          | Moderate-High |
| SOC analyst (tier 1)               | 6.1  | 6.8  | 7.4       | +1.3          | High          |

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, entry-level software development scores 8.1, entry data analysis 8.2, and licensed engineering 4.0.

**Read the table this way:** nothing in cybersecurity scores below 5.0. It genuinely is the best position in technology, and it is a mid-table position in this index overall. Your job, from here, is to move up the table toward the roles that face the adversary — not to hope the table moves.

# Why cybersecurity is protected — the six factors



Here's how incident response and threat hunting answers them. Knowing *why* the entry SOC rung scores 7.4 while incident response holds 5.0 is what tells you which way to move.

## HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

**Most of tier-one triage, and very little above it.**

Alert triage and false-positive filtering. Log correlation and enrichment. Vulnerability scanning and reporting. Standard playbook execution. Compliance evidence collection. Phishing detection and classification.

**And the projection here is unusually specific.** Gartner projects that more than 50% of SOC Tier 1 analyst responsibilities will be handled by AI by 2028. That is a named forecast about the exact role most people are told to start in.

What resists: deciding whether this is an incident or noise, working out what an adversary is actually trying to do, judgment when the attack has no precedent, making the call to shut down production, and explaining a breach to a board or a regulator. Those are the things to be building toward.

## HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

**Harder than the marketing suggests, and the section on the 4.8 million figure covers why in detail.**

The short version: the shortage is for experienced people. Practitioners consistently report that "entry level" listings ask for two years of experience, and ISC2's 2025 workforce study found budget constraints have overtaken talent shortage as the number one hiring barrier.

**The way through isn't to wait for the entry rung to soften — it may not.** It's to enter *beside* it: build demonstrable, hands-on evidence — home labs, capture-the-flag competitions, bug bounty work — because in a field with no license that is the credential; aim at incident response, security engineering or offensive work rather than pure SOC monitoring; and use apprenticeship or degree-apprenticeship routes, which are paid and solve the experience paradox directly. Those are concrete moves, not a hope, and the career-hunting checklist later in this guide lays them out.

**DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)**

**Almost none of it.** Security work is fully digital, which removes one protection entirely.

**DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)**

**At the incident and architecture level, substantially.**

Someone has to make the call to take a system offline, and someone has to answer to a board or a regulator afterwards. That accountability attaches to a person — which means the fastest way to become protected is to get closer to the seat that carries it.

**DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)**

**Weakly.** There is no cybersecurity license, though regulatory regimes increasingly require named accountable individuals for security posture — which is a partial and growing protection.

**HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)**

**We rate this 9.0, and the section on adversarial novelty explains why cybersecurity's novelty is a different kind from every other career in this index.** Moving toward the work that actually meets that novelty is the whole direction of travel from here.

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## The 4.8 million number, and what it actually means

This is the most useful section in the guide, because the claim it examines is everywhere.

**The numbers are real:**

- ISC2's Workforce Study puts the global gap at roughly **4.8 million unfilled positions**
- CyberSeek — a joint effort by CompTIA, Lightcast and NIST — puts the US figure at around **700,000–750,000**
- The World Economic Forum estimates the workforce needs to grow **87%** to meet demand

- BLS projects information security analyst employment growing 29–33% through 2034, roughly seven times the average across all occupations

**Every one of those is accurate. And a beginner should read them carefully.**

**First: the shortage is for experience, not for people.** The consistent report from practitioners and hiring data alike is that "entry level" postings ask for two years of experience. Organisations want people who can direct and challenge security work, not people who need training. The gap persists precisely because new entrants aren't being developed.

**Second: budget has overtaken talent as the constraint.** ISC2's 2025 study found budget cuts displacing talent shortage as the primary hiring barrier. A vacancy that exists on paper but has no funded headcount behind it does not employ anyone.

**Third, and most directly relevant to our score:** the entry route being marketed is the one being automated. Gartner's projection that more than half of SOC Tier 1 responsibilities will be AI-handled by 2028 lands squarely on the job that bootcamps, certifications and university programs point graduates toward.

**Fourth, the pipeline problem this creates.** One industry analysis puts it well: a workforce with no entry-level pipeline produces no tier-two analysts and no senior generalists in five years. The same argument we found in law — the sector may be making a self-defeating mistake — applies here, and nobody has resolved it.

**Where this leaves you.** Cybersecurity is a genuinely good career with genuine demand. It is not the frictionless six-figure on-ramp it is marketed as, and the specific route being sold is the weakest one. The repositioning section covers what to do about that from where you stand.

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## Adversarial novelty — a protection nothing else in this index has

We rate threat hunting 9.0 on novelty, and it is worth understanding why, because the mechanism is unique.

AI is strong on patterns and weak on genuine novelty. Most careers earn a high rating here through complexity — a one-off engineering site, an atypical patient, a case with no precedent.

**Cybersecurity earns it because someone is deliberately manufacturing novelty against you.**

That distinction matters more than it sounds. An unusual medical presentation is unusual by accident. An unusual attack is unusual on purpose — designed specifically to look benign, or to

exploit the gap between what a detection system was trained on and what is actually happening.

This creates a property no other profession in this index has: **the novelty is adaptive**. Whatever pattern a defensive system learns to recognize, an adversary has an incentive to stop matching it. There is no equilibrium where the patterns settle down and the work becomes routine, because the other side is paid to prevent exactly that.

That is why threat hunting holds 5.0 despite no physical component, no license, and a fully digital workflow. Almost nothing else in this index survives that combination of missing protections.

**But notice where the protection does not reach.** A tier-one SOC analyst is not facing adversarial novelty. They are working a queue of alerts against a playbook — pattern-matching, in a role explicitly designed to be pattern-matching. That is why it rates 4.5 rather than 9.0, and why it scores 7.4.

The general lesson: adversarial work resists automation better than merely complex work. Complexity can be learned. An opponent who changes when you learn cannot be. Any career with a genuine adversary — security, litigation, fraud investigation, negotiation, competitive strategy — carries a protection that complexity alone does not provide.

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## What the trend shows

**SOC analyst tier 1:** 6.1 → 6.8 → 7.4. Up 1.3.

**Incident response:** 4.2 → 4.6 → 5.0. Up 0.8.

**The gap widened from 1.9 points to 2.4.** The distinctive feature here is that both ends moved, and the protected end moved faster than protected ends usually do — because cybersecurity has no license or physical requirement holding it still, only judgment and adversarial dynamics.

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# How durable is the protection?

| PROTECTION                   | STRENGTH TODAY         | DURABILITY           | WHERE THE PRESSURE IS COMING FROM                                        |
|------------------------------|------------------------|----------------------|--------------------------------------------------------------------------|
| Judgment under novelty       | Strongest available    | Most durable here    | Adaptive adversaries prevent patterns from settling. Uniquely resistant. |
| Human trust / accountability | Real at incident level | Durable              | Someone must decide to take production down and answer for it.           |
| Regulatory / licensing       | Weak but growing       | Strengthening slowly | Regulation increasingly requires named accountable individuals.          |
| Physical / embodied          | None                   | n/a                  | Fully digital work.                                                      |

**The honest read.** Cybersecurity's protection is unusual: it comes almost entirely from the adversary rather than from structure. That is genuinely durable — as long as the adversary exists, which is not in doubt — but it protects the roles that face the adversary directly, and not the ones processing alerts behind them, which is why the direction you steer matters more than where you happen to start.

# How the role will actually change

| WAS THE JOB                                   | IS BECOMING THE JOB                         |
|-----------------------------------------------|---------------------------------------------|
| Triaging the alert queue                      | Reviewing what the system escalated         |
| Correlating logs by hand                      | Interrogating automated correlation         |
| Running the vulnerability scan                | Judging which findings actually matter here |
| Following the playbook                        | Deciding when the playbook doesn't apply    |
| Learning the environment by working the queue | Learning it... some other way, unresolved   |
| Deciding to shut down production              | Unchanged                                   |

**The fifth row is the recurring problem across this index**, and cybersecurity has it acutely. Analysts learned an organisation's environment — what normal looks like — by working



thousands of alerts. That was tedious, and it was how you developed the instinct to know when something is wrong. Nobody has replaced it — which means the people who go out of their way to learn an environment deeply, rather than just clear its queue, have an edge that's about to get scarcer.

**And there's a second-order effect worth naming.** AI improves both sides simultaneously. This is the only career we score where the same technology strengthens the attacker and the defender at once, which is part of why demand holds even as tier-one work automates.

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## Human strengths that will matter most

1. **Adversarial thinking** — modeling what someone actively hostile would do. The core skill, and the hardest to teach. 2. **Deciding under pressure with incomplete information** — most incidents are ambiguous at the moment the decision has to be made. 3. **Accountability** — being the person who takes production down, and answers for it. 4. **Explaining to non-technical people** — boards, regulators, executives. Increasingly the difference between a good analyst and a security leader. 5. **Verification** — knowing when an automated detection is wrong in either direction.

Notably absent: tool fluency, alert throughput and certification breadth. These were the traditional markers of a strong junior and are the fastest-depreciating. If your current value is mostly on that second list, the work is to shift it onto the first.

And underneath all five, two traits the score can't grade: **curiosity and adaptability**. The specific tools — the SIEM, the EDR platform, the framework of the moment — will turn over many times across a career, and so will the attacks themselves. The people who compound are the ones who stay genuinely curious about how systems break and re-tool fastest, rather than defending the one stack they learned. In a field with no license and a permanently moving adversary, that flexibility is itself a form of protection — and increasingly it's what employers are hiring for: the trajectory toward judgment and investigation, not the current certification count.

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# Other things to consider about this job

## What's genuinely good about it

**Demand is real and structurally driven.** BLS projects 29–33% growth through 2034 — roughly seven times the average. Ransomware, breach frequency and regulatory obligations are all increasing, and none of those is cyclical.

**Entry pay is solid:** SOC and GRC analyst roles report roughly \$70,000–\$85,000 depending on location, junior incident response \$70,000–\$90,000, and junior penetration testing \$75,000–\$95,000.

**Entry routes are unusually flexible.** ISC2's hiring research found around 90% of organisations will consider candidates with only IT experience, and 89% accept entry-level certifications in place of a degree. Very few professional fields are that open.

**What practitioners report as rewarding:** the work genuinely matters when it matters; there is a real opponent, which some people find energizing in a way ordinary problem-solving isn't; continuous learning is intrinsic rather than optional; and progression to well-paid senior and leadership roles is fast for people who stay.

## What's genuinely hard about it

**Burnout is severe and unusually well documented.**

- Bitsight's 2025 survey of over 1,000 risk and security professionals found 47% reporting some level of burnout
- Proofpoint's Voice of the CISO 2025 found 63% of CISOs had experienced or witnessed burnout in the past year
- Sophos measured 4.8 hours per week lost to burnout, up more than 25% year on year
- Splunk's State of Security, surveying 2,058 practitioners, found 52% of security teams overworked
- Industry analysis finds attrition exceeds other technology roles

**On-call is a permanent feature,** not an early-career phase. Incidents happen at night and at weekends.

**The certification treadmill is real** and never stops, and much of it is self-funded.

**And the entry paradox is the practical obstacle:** the field wants experience, and the traditional way of getting it is being automated.

## Who this work suits — and who it doesn't

Cybersecurity tends to suit people who:

- **Think adversarially** — instinctively asking how something could be abused rather than how it should work. The defining trait, and largely untrainable.
- **Stay calm when things are going badly** — incidents are the job
- **Are comfortable with ambiguity** and deciding without complete information
- **Enjoy continuous learning** rather than tolerating it
- **Can explain technical things to people who don't want the detail** — this is what separates analysts from leaders

It's harder for people who:

- Need **predictable hours** — on-call is structural
- Want **defined, completable tasks** — security is never finished
- Find **sustained pressure** corrosive rather than energizing
- Chose it because it was marketed as the **safe tech career**. That framing is what the 4.8 million section is about.

If some of that second list sounds like you, it isn't a reason to abandon the field — it's a reason to steer toward the parts of it that fit and toward the protected tracks, which is what the rest of this does.

## What else is moving this market

**Regulation is a genuine tailwind.** Compliance obligations create demand that has nothing to do with attack volume, and increasingly require named accountable individuals — which is a protection our regulatory factor is only beginning to reflect.

**Retention is a bigger drain than layoffs.** ISACA's research found organisations struggling to retain security staff even without cutting — voluntary turnover driven by burnout is a larger problem than involuntary loss, which is why senior people remain scarce.

**And AI is being adopted cautiously.** Splunk found 33% of teams planning to fill skills gaps with AI and automation, but only 11% trusting AI completely for mission-critical tasks. That gap between adoption and trust is where the human roles sit.

# The AI-native advantage — what it actually looks like

**The situation:** security teams are adopting AI faster than they trust it. Only about one in nine trusts it completely for mission-critical work, while a third plan to use it to fill skills gaps. That gap is the opportunity. Being already in it, close to the work, is a real advantage here — you can build this fluency inside a home lab, a live internship or a first job, which is exactly where it counts.

## WHAT AN AI-NATIVE SECURITY PROFESSIONAL LOOKS LIKE

- 1. Verify.** Knowing when an automated detection is wrong — in either direction. A false negative is a breach; a false positive at scale is alert fatigue, which is itself a security failure.
- 2. Think adversarially about AI itself.** Attackers are using the same tools. Understanding how to attack and defend AI systems is a specialism that barely existed three years ago and is growing fast.
- 3. Move up the escalation chain deliberately.** Tier one is automating. Every step toward investigation, hunting and architecture moves someone from a 7.4 to a 5.0.
- 4. Own a decision.** Everything in this guide points at accountability — being the person who calls it.
- 5. Communicate.** Explaining an incident to a board is the skill that turns an analyst into a leader, and it is chronically undersupplied.

All five are available to you now — not after some future certification. Turning them into concrete moves is the rest of this guide.

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## Where a cybersecurity career can lead later

Broad and well-paid: incident response and digital forensics, security architecture, offensive security and red teaming, cloud and application security, AI/ML security, governance risk and compliance, security leadership and CISO roles, and consultancy.

**The pattern in this index holds, with a security-specific twist:** roles facing the adversary directly score better than roles processing what the adversary triggered. Cloud security, AI/ML defense and application security are named consistently as the most in-demand skills, and all three sit on the protected side of that line. That line is the compass for every choice below.

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# Repositioning from here

The program-selection stage is behind you. The useful questions now are different — and the good news bears repeating: **almost none of your exposure is fixed yet.** Cybersecurity is a 2.4-point spread from 5.0 to 7.4, and where you land inside it is decided mostly by things still in front of you. Nearly all of them point the same way: toward the roles that face the adversary directly.

## If you're still in the degree or training

Three levers, in order of leverage.

**1. The specialization is the whole game.** Tier-one SOC monitoring (7.4) is the exposed rung and the one Gartner projects half-automated by 2028. The protected destinations — incident response and threat hunting (5.0), security architecture (5.5), penetration testing and offensive work (5.7) — are reached by concentration, not by generic security coursework. If you haven't committed to a track, this is the single highest-leverage decision left. If you have, hold it against the scores above and ask honestly whether it points at *investigation and judgment* or at *queue-work*.

**2. Hands-on evidence, while it's still free to build.** In a field with no license, demonstrable capability is the credential — and it's the one thing you can accumulate right now for nothing. Home labs, capture-the-flag competitions, bug bounty work, contributing to open-source security tooling: **every hour of it is a step from the exposed end toward the protected one**, and it's far cheaper to acquire as a student than to manufacture later. Certification breadth alone is the fastest-depreciating asset here; verifiable, hands-on capability is not.

**3. The apprenticeship, and the experience it generates.** The single hardest problem in this field is that "entry level" wants two years of experience. Degree apprenticeships and placement routes are paid, and they solve that paradox directly — they are the on-ramp the marketed route doesn't give you. Secure one if you possibly can, and aim it at a protected track rather than pure SOC monitoring.

## If you've graduated and you're job-hunting

Here the title on the offer matters more than it looks, because the most common early-career security title is also the most exposed.

- **"Tier-one SOC analyst" is the 7.4.** It's the default graduate destination and the automating layer at once — the exact role Gartner projects half-automated by 2028. Taking one isn't fatal — but take it as a deliberate two-year staging post with a move to tier two and beyond in mind, not as the plan.

- **Aim at the roles that face the adversary (5.0–5.7).** Incident response and threat hunting, security engineering and architecture, offensive work — more protected, and the place the adversarial-novelty shield actually reaches.
- **Attach to the in-demand protected skills.** Cloud security, AI/ML defense and application security are named consistently as the most in-demand, and all three sit on the protected side of the line. "Open to any security role" aims straight at the exposed rung.
- **Chase the seat that carries a decision.** Anything where you investigate, hunt, architect or own the call to take production down moves you up the scale — even at a smaller organisation, where that responsibility tends to arrive years sooner.

## The AI-native move, from where you stand

You can't undo the tasks that are already automating, but you can be the entrant who directs these tools rather than competes with them. The five capabilities above — verify, think adversarially about AI itself, move up the escalation chain, own a decision, communicate — are all available right now, inside a home lab, a current internship or a first job. The most powerful for someone already in it is verification: be the person who knows when an automated detection is wrong in either direction — a false negative is a breach, a false positive at scale is alert fatigue, and both are security failures. In a field with no license, that judgment is the credential, and you can start building it this term, not after some future certification.

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## The career-hunting checklist

The analysis tells you where the exposure is. This is what to do about it — the part that's actually in your hands. Most of the protection in cybersecurity is built here, in these habits and choices, rather than handed to you. The first list is for anyone still in the degree or training; the second is for the job hunt itself.

### During the degree / training — build what can't be automated

- **Aim at a protected specialization, not the SOC queue.** Incident response and threat hunting (5.0), security architecture (5.5) and offensive security / penetration testing (5.7) are the protected end; tier-one SOC monitoring (7.4) is the exposed one. Point every elective and placement at the former.
- **Build hands-on evidence before anyone asks for it.** Home labs, capture-the-flag competitions, bug bounty work, contributing to open-source security tooling. In a field



with no license, demonstrable capability is the credential — and it's free to a sixteen-year-old tonight.

- **Get Security+ or an equivalent early, then specialize.** The baseline cert opens the door; depth in one protected track is what carries you through it. Certification breadth alone is among the fastest-depreciating things you can collect.
- **Chase the apprenticeship or placement route.** Degree apprenticeships are paid and solve the experience paradox directly — the single hardest problem in this field — and they're badly under-used.
- **Get fluent with AI as a defender, not a shortcut.** Learn to verify an automated detection in both directions, and learn how to attack and defend AI systems themselves — a specialism that barely existed three years ago.
- **Feed curiosity and adaptability on purpose.** The tools and the attacks both turn over constantly; the person who re-tools fastest, and stays genuinely interested in how systems break, compounds.
- **Keep the frame wide.** Vet computer science with a security specialization, apprenticeship routes, and adjacent technical paths properly before committing. The most crowded lane is the SOC-tier-one bootcamp, and it points at the automating role.

## During the job hunt — aim, don't just apply

- **Target the protected lanes on purpose.** Incident response, security engineering, offensive work, cloud and application security — not "SOC analyst, open to anything," which points straight at the 7.4.
- **Read the title, not just the offer.** "Tier-one SOC analyst" is the automating layer. If you take one, take it as a deliberate two-year staging post with a move to tier two and beyond in mind, not as the plan.
- **Don't grab a role just because it's on offer in an exposed lane.** A role being available doesn't make it safe. An employer still running a large tier-one SOC intake may simply be behind the curve — and that's exactly the rung Gartner projects half-automated by 2028. Availability can be a lagging indicator, not a green light.
- **Prefer the seat that faces the adversary.** A role where you investigate, hunt or architect — where someone is deliberately manufacturing novelty against you — carries the protection nothing else in this index has. Proximity to the adversary is proximity to the protection.
- **Ask the employer where AI is heading in the role.** Whether they're building the job around investigation and verification, or waiting to be forced to, tells you if the seat grows or is quietly being hollowed out.

- **Weigh trajectory over the starting salary.** Entry pay is solid across the field — SOC and GRC \$70,000–\$85,000, junior incident response \$70,000–\$90,000, junior penetration testing \$75,000–\$95,000 — so choose the lane that ages best, not the highest first-year number in an automating one.

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## Go deeper — further reading

- **Information Security Analysts — Occupational Outlook Handbook** — US Bureau of Labor Statistics. The primary source for the growth projection and wage data. Free, and the least commercially interested source available on this career.
- **ISC2 Cybersecurity Workforce Study** — the origin of the 4.8 million figure, and worth reading in full rather than via the headline. The 2025 edition's finding that budget constraints have overtaken talent shortage as the top hiring barrier is the part almost nobody quotes.
- **CyberSeek** — CompTIA, Lightcast and NIST. An interactive map of actual US cybersecurity demand by location and role. Useful for seeing which roles are genuinely open near you rather than in aggregate.

### The counterargument — read this one:

The strongest case against our 7.4 for tier-one work is the demand data itself: 4.8 million unfilled positions globally, 29–33% projected growth, and organisations so short-staffed that 90% will consider candidates with only IT experience. On that reading, automating tier-one triage doesn't eliminate the role — it makes an impossibly under-resourced function survivable, and the analysts move up rather than out.

There's a version of this that is explicitly the industry's plan: redesign entry-level roles around AI augmentation rather than eliminating them, precisely because a workforce with no entry pipeline produces no senior generalists in five years.

**Where we land:** we think the demand is real and the pipeline argument is sound — and that the same argument was made loudly in law for two years while graduate hiring moved the other way. Intent and behavior are different things. We score the market as it is.

**The specific thing we're watching:** whether tier-one SOC postings hold or fall as the Gartner 2028 horizon approaches. If organisations genuinely redesign entry roles around augmentation rather than reducing them, our 7.4 is too high and we will lower it.



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## Bottom line

**Cybersecurity is the best position in technology and a mid-table position overall — and the gap between those two statements is where most of the confusion lives.** That's the whole reason being already in it is not a problem: the decisive moves were always going to happen from here.

Its protection is genuinely unusual: **an adaptive adversary means the patterns never settle, which is a shield nothing else in this index has.** That's why threat hunting holds 5.0 despite having no license, no physical requirement and a fully digital workflow.

**But the marketing and the hiring reality have come apart.** The 4.8 million figure is real and it describes a shortage of *experience*, not of people. Budget constraints have overtaken talent as the primary barrier. And Gartner projects more than half of tier-one SOC work handled by AI by 2028 — which is precisely the role that bootcamps, certifications and degree programs point graduates toward. Standing still on that rung is the one thing not to do.

**So the useful advice isn't a verdict on the field — it's a direction.** Build hands-on evidence, because in a field with no license that is the credential and it's free to acquire. Aim at incident response, security engineering or offensive work rather than SOC monitoring. Take a SOC role if it's the way in, but treat it as a two-year staging post rather than a destination. And build real AI fluency — verification above all — inside whatever role or lab you're in now, because that's the differentiator this field rewards fastest.

And weigh the burnout data honestly: 47% reporting burnout and attrition above other technology roles is not a footnote. But one thing worth saying plainly. **Cybersecurity is one of very few careers where an adversary guarantees the work stays interesting, and where you can start building genuine, verifiable capability tonight for nothing.** That combination is rare, and it's still available to you from exactly where you stand.

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## Questions worth sitting with

*Whether you're the one in it or a parent thinking it through, these are the conversations that move the needle now. Read "you" as whoever the decision is about.*

## Where you actually stand

1. Which track are you on — or aiming at — and does it point at *investigation and judgment* (incident response, threat hunting, architecture) or at *queue-work* (tier-one SOC monitoring)? 2. Looking at the score table, which destination are you currently pointed at, and is it the one you'd choose on purpose? 3. Do you have any demonstrable, hands-on evidence yet — a home lab, capture-the-flag results, a bug bounty, an open-source contribution — or only coursework? If it's the latter, what's the nearest way to change that?

## The move from here

4. What's the single most protected role you could realistically aim for from where you are now, and what would make you a credible candidate for it? 5. If you're still studying: what's the most valuable placement or apprenticeship still open to you, and is there a reason not to chase it? 6. If you're job-hunting: are you aiming at a protected lane on purpose, or applying to whatever's open? Name the three roles you're actually chasing.

## The AI question, practically

7. Of the five capabilities — verify, think adversarially about AI itself, move up the escalation chain, own a decision, communicate — which one could you visibly demonstrate within the next few months? 8. Can you point to a moment where you caught an automated detection that was wrong — in either direction? If not, where's the easiest place to start building that skill?

## Testing it against reality

9. **The most valuable single action, available tonight and free:** set up a home lab, or start capture-the-flag competitions. Nothing else demonstrates capability as directly, and almost nobody at the start of a career has any. 10. If the first offer that comes is a tier-one SOC role, what's your plan — take it as a deliberate two-year staging post with an exit toward tier two in mind, or hold out for something closer to investigation? Decide before the offer, not after.

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*This is analysis and informed judgment about a fast-moving situation. For cybersecurity specifically, we've flagged that widely-quoted workforce shortage figures describe a shortage of experience rather than of people, and that the entry route most commonly marketed is the one most directly affected by automation.*

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

# Technical scoring appendix — Cybersecurity

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

## 1. The formula

$$\begin{aligned}\text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment})\end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

**Bands.** Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

## 2. Rating anchors

### HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

**Task automatability (A)** — *what share of the working week could a capable current system already do to an acceptable standard?*

| RATING | ANCHOR                                                                    |
|--------|---------------------------------------------------------------------------|
| 0–2    | Almost nothing. The work is physical, relational or wholly novel.         |
| 3–4    | Administrative surround only — notes, scheduling, correspondence.         |
| 5–6    | A meaningful minority of the week, mostly documentation and lookup.       |
| 7–8    | A large share, including tasks central to the junior version of the role. |
| 9–10   | Most of the described duties, at or near professional standard.           |

**Entry-path erosion (E)** — *how much has the traditional route into this work narrowed?*

| RATING | ANCHOR                                                                          |
|--------|---------------------------------------------------------------------------------|
| 0–2    | Protected by statute or physical necessity. Training hours cannot be automated. |
| 3–4    | Stable. Some junior tasks absorbed; the apprenticeship survives.                |
| 5–6    | Visible narrowing. Employers prefer experience; junior intake reduced.          |
| 7–8    | Substantial. The work juniors learned on has largely gone.                      |
| 9–10   | The training layer and the automatable layer were the same layer.               |

**Physical / embodied (P)** — *must this be done in person, with a body, in an unpredictable setting?*

| RATING | ANCHOR                                                       |
|--------|--------------------------------------------------------------|
| 0–2    | Fully screen-based. Location-independent.                    |
| 3–4    | Occasional presence; the core work is not physical.          |
| 5–6    | Regular physical presence, in largely controlled conditions. |
| 7–8    | Substantially physical, with some variability of setting.    |
| 9–10   | Irreducibly physical, in conditions that differ every time.  |

**Human trust / accountability (T)** — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

| RATING | ANCHOR                                                              |
|--------|---------------------------------------------------------------------|
| 0–2    | Output is anonymous and reviewed by others.                         |
| 3–4    | Work is signed off by someone else. No personal reliance.           |
| 5–6    | Some client relationship; accountability is institutional.          |
| 7–8    | Named responsibility, with commercial or professional consequence.  |
| 9–10   | The relationship is the service, and liability attaches personally. |

**Regulatory / licensing (R)** — *does the law reserve this act to a qualified human?*

| RATING | ANCHOR                                                                |
|--------|-----------------------------------------------------------------------|
| 0–2    | No license exists and none is proposed.                               |
| 3–4    | Certification is customary but not reserved.                          |
| 5–6    | Partial reservation, varying by jurisdiction or task.                 |
| 7–8    | Licensed practice, with some unreserved adjacent work.                |
| 9–10   | Comprehensive statutory reservation, including of the training route. |

**Judgment under novelty (J)** — *how often does the work present something with no matching precedent, where being wrong is costly?*

| RATING | ANCHOR                                                           |
|--------|------------------------------------------------------------------|
| 0–2    | Routine by design. Deviation is an error, not a case.            |
| 3–4    | Occasional exceptions, handled by escalation.                    |
| 5–6    | Regular non-standard cases within a known range.                 |
| 7–8    | Frequent genuine novelty with material consequences.             |
| 9–10   | Novelty is constant, and in some fields adversarially generated. |

### 3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

#### INCIDENT RESPONSE / THREAT HUNTING — 5.0 (MODERATE)

| FACTOR                       | WEIGHT | 2023 | 2025 | 2026         | CONTRIBUTION TO 2026 SCORE |
|------------------------------|--------|------|------|--------------|----------------------------|
| Task automatability          | 35%    | 3.2  | 4.1  | 5.1          | 1.78                       |
| Entry-path erosion           | 15%    | 4.0  | 4.5  | 5.0          | 0.75                       |
| Physical / embodied          | 15%    | 1.0  | 1.0  | 1.0          | 1.35                       |
| Human trust / accountability | 15%    | 7.5  | 7.5  | 7.5          | 0.38                       |
| Regulatory / licensing       | 10%    | 3.5  | 3.5  | 3.5          | 0.65                       |
| Judgment under novelty       | 10%    | 9.0  | 9.0  | 9.0          | 0.1                        |
|                              |        |      |      | <b>Total</b> | <b>5.01 → 5.0</b>          |

#### SECURITY ARCHITECTURE — 5.5 (MODERATE-HIGH)

| FACTOR                       | WEIGHT | 2023 | 2025 | 2026         | CONTRIBUTION TO 2026 SCORE |
|------------------------------|--------|------|------|--------------|----------------------------|
| Task automatability          | 35%    | 4.3  | 5.2  | 6.1          | 2.13                       |
| Entry-path erosion           | 15%    | 4.0  | 4.5  | 5.0          | 0.75                       |
| Physical / embodied          | 15%    | 1.0  | 1.0  | 1.0          | 1.35                       |
| Human trust / accountability | 15%    | 7.0  | 7.0  | 7.0          | 0.45                       |
| Regulatory / licensing       | 10%    | 3.5  | 3.5  | 3.5          | 0.65                       |
| Judgment under novelty       | 10%    | 8.5  | 8.5  | 8.5          | 0.15                       |
|                              |        |      |      | <b>Total</b> | <b>5.49 → 5.5</b>          |



## PENETRATION TESTING / OFFENSIVE — 5.7 (MODERATE-HIGH)

| FACTOR                       | WEIGHT | 2023 | 2025 | 2026         | CONTRIBUTION TO 2026 SCORE |
|------------------------------|--------|------|------|--------------|----------------------------|
| Task automatability          | 35%    | 4.6  | 5.5  | 6.5          | 2.27                       |
| Entry-path erosion           | 15%    | 4.2  | 4.8  | 5.2          | 0.78                       |
| Physical / embodied          | 15%    | 1.5  | 1.5  | 1.5          | 1.27                       |
| Human trust / accountability | 15%    | 6.5  | 6.5  | 6.5          | 0.53                       |
| Regulatory / licensing       | 10%    | 3.0  | 3.0  | 3.0          | 0.7                        |
| Judgment under novelty       | 10%    | 8.5  | 8.5  | 8.5          | 0.15                       |
|                              |        |      |      | <b>Total</b> | <b>5.71 → 5.7</b>          |

## GRC / COMPLIANCE — 6.1 (MODERATE-HIGH)

| FACTOR                       | WEIGHT | 2023 | 2025 | 2026         | CONTRIBUTION TO 2026 SCORE |
|------------------------------|--------|------|------|--------------|----------------------------|
| Task automatability          | 35%    | 4.6  | 5.8  | 6.7          | 2.34                       |
| Entry-path erosion           | 15%    | 4.5  | 5.2  | 5.8          | 0.87                       |
| Physical / embodied          | 15%    | 1.0  | 1.0  | 1.0          | 1.35                       |
| Human trust / accountability | 15%    | 6.0  | 6.0  | 6.0          | 0.6                        |
| Regulatory / licensing       | 10%    | 5.0  | 5.0  | 5.0          | 0.5                        |
| Judgment under novelty       | 10%    | 5.5  | 5.5  | 5.5          | 0.45                       |
|                              |        |      |      | <b>Total</b> | <b>6.11 → 6.1</b>          |

SOC ANALYST (TIER 1) — 7.4 (HIGH)

| FACTOR                       | WEIGHT | 2023 | 2025 | 2026  | CONTRIBUTION TO 2026 SCORE |
|------------------------------|--------|------|------|-------|----------------------------|
| Task automatability          | 35%    | 5.3  | 6.9  | 8.3   | 2.91                       |
| Entry-path erosion           | 15%    | 5.5  | 6.5  | 7.2   | 1.08                       |
| Physical / embodied          | 15%    | 1.0  | 1.0  | 1.0   | 1.35                       |
| Human trust / accountability | 15%    | 4.5  | 4.5  | 4.5   | 0.82                       |
| Regulatory / licensing       | 10%    | 3.0  | 3.0  | 3.0   | 0.7                        |
| Judgment under novelty       | 10%    | 4.5  | 4.5  | 4.5   | 0.55                       |
|                              |        |      |      | Total | 7.41 → 7.4                 |

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Incident response / threat hunting

| EDITION   | A   | E   | PROTECTION DEFICIT (FIXED) | SCORE |
|-----------|-----|-----|----------------------------|-------|
| 2023      | 3.2 | 4.0 | 2.48                       | 4.2   |
| 2025      | 4.1 | 4.5 | 2.48                       | 4.6   |
| Fall 2026 | 5.1 | 5.0 | 2.48                       | 5.0   |

**Movement decomposition:** automatability contributed +0.66 and entry-path erosion +0.15, for a total of +0.8 points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

## 5. Sensitivity

How far the score moves if a single factor is rated one point differently:

| FACTOR                       | ±1 POINT MOVES THE SCORE BY |
|------------------------------|-----------------------------|
| Task automatability          | ±0.35                       |
| Entry-path erosion           | ±0.15                       |
| Physical / embodied          | ±0.15                       |
| Human trust / accountability | ±0.15                       |
| Regulatory / licensing       | ±0.10                       |
| Judgment under novelty       | ±0.10                       |

**What this means in practice.** A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

## 6. Stated limitations

**We measure capability, not deployment.** Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

**We do not measure demand.** A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

**We do not measure oversupply.** The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

**We do not measure industry economics.** A profession can contract for reasons entirely unrelated to technology.

**We do not measure whether the work is worth doing.** Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

**Sub-track definitions are ours.** Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

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## 7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

**Across the index:**

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

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Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.